Int. J. Rock Mech. Min. Sci. & Geomech. Abstr. Vol. 13, pp. 103–112. Pergamon Press 1976. Printed in Great Britain

Development of Stress-Induced Microcracks in Westerly Granite

PAUL TAPPONNIER*
W. F. BRACE†

Ion-thinned samples under the scanning electron microscope reveal new details of the development of dilatant microcracks in stressed Westerly granite at 500 bars confining pressure, room temperature. Most cracks caused by laboratory-induced stress are long, straight and narrow, with sharp ends, in contrast to natural cavities, which are blunt, bridged and discontinuous. New cracks first begin to appear within about 500 bars stress of C′, at grain boundaries and healed transgranular cracks. At around 75% of the peak stress, new transgranular cracks form, starting particularly at high angle interfaces of dissimilar minerals or, rarely, at inclined pre-existing shear cracks. Peak stress coincides with kinking in suitably oriented biotite, which together with magnetite appear to have been the centers of particularly intense brittle cracking. Crack density approximately doubles up to peak stress. The scarcity of shear cracks raises doubts about the usual frictional interpretation of hysteresis in dilatancy; hysteresis could originate at interface-related cracks or at axial cracks themselves.

INTRODUCTION

We recently described a new method for observing the minute cracks and pores in crystalline rocks [1, 2]. The principal new feature of this method was the use of *ion thinning* to remove damaged material introduced by sawing or polishing of a section. Ion thinning exposes material virtually undamaged down to atomic dimensions [2]. Examination of such a thinned section with a scanning electron microscope (SEM) reveals details of cracks and pores with a resolution of about 10^{-6} cm.

One application of our new method is to study stress-induced cracking [2, 3], particularly that associated with applied nonhydrostatic compressive stress. There is currently great interest in this process, particularly with regard to dilatancy [4] and its possible role in earthquake prediction [5, 6]. The previous studies of dilatant cracking have been of two types: (a) those in which crack characteristics are inferred from electrical or elastic measurements made under stress [7–10]; or (b) those in which optical examination is carried out after stress is removed [11–15]. The former suffer in that only some averaged crack property is obtainable, the latter in that many cracks are below optical resolution [2]. The present method seemed promising on both grounds. Shape and location of individual cracks is apparent in the SEM so that a much more detailed characterization of dilatant microcracks should be possible and resolution is extended nearly a hundred times over optical methods.

A number of important aspects of dilatant cracking remain unresolved: (a) where and how do cracks first form; (b) does their first appearance coincide with so-called C', the stress at which dilatancy is first externally detected; (c) do actual cracks have the forms suggested by strain measurements [4]; (d) if there is more than one type of crack, do certain types coincide with certain regions of the complete stress–strain curve [13]; (e) what is the distribution of crack lengths and aspect ratios; (f) how are dilatant cracks related to faults? We have studied a number of ion-thinned sections with the SEM in an attempt to resolve some of these questions, for at least one well-characterized rock, Westerly granite [4]. Here, we will restrict attention principally to crack development in this rock up to the ultimate or peak stress.

EXPERIMENTAL PROCEDURE

Ideally, crack development should be studied for a single sample deformed to successively higher stresses, as in our previous study of the effects of high pressure and temperature on natural cracks [3]. Techniques are not yet available to do this for stress, however, so we followed crack development in a suite of identical samples, all cored from the same block and deformed at the same pressure, temperature and strain rate. In so doing we relied on the reproducible behavior of Westerly granite; stress–strain behavior and ultimate stress of this rock typically agree closely from sample to sample [13, 16].

* Laboratoire de Géologie Structurale, University of Montpellier, France.
† Department of Earth and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, U.S.A.

R.M.M.S. 13/4—A

TABLE 1. SAMPLES STUDIED. THE T SERIES ARE NEW TO THIS STUDY; THE W SERIES ARE FROM [13]

Number	P bars	Stress kbars	Region
T0	0	0	Below C', Figure 1
T1	500	1	Below C', Figure 1
T2	500	2	Below C', Figure 1
T3	500	3	Above C', Figure 1
T4	500	4.5	Above C', Figure 1
T5	500	5.3	Close to peak stress
W1	35	-	Many en echelon cracks, nearly complete fault, incipient slabbing
W2	300	-	Extensive cracking, incipient faulting
W3	800	-	Extensive transgranular cracking
W4	800	-	Completely faulted, conspicuous unloading cracks
W5	1500	-	Moderate cracking along axial and inclined directions

The choice of confining pressure, P, for our suite of samples was governed by two factors. First, we wished to do a confined experiment to avoid the difficulties usually associated with end effects in unconfined experiments [16]. But there was also the damaging effects of pressure alone: a pressure cycle of a kbar or more produced local but marked damage of many of the natural cavities in Westerly granite [3]. Since we wished to observe stress and not pressure effects, pressure was kept as low as possible. As a compromise, we chose 500 bars; temperature was 23°C and strain rate 10^{-5} sec^{-1}.

One sample, T5, was taken to as near peak stress as possible without faulting, and its stress–volumetric strain curve is shown in Fig. 1. One unstressed sample, TO, was retained for reference, and the remaining samples (Table 1) were stressed to intermediate levels, bracketing C' for this rock which at this pressure is about 2·5 kbar. Only in T5 were traces of visible cracking evident; these took the form of tiny incipient faults near one end of the sample.

We also re-examined several samples of Westerly granite which were stressed by Wawersik [13] to beyond the peak stress. By using special unloading techniques, these were recovered intact, that is, without having suffered the catastrophic failure observed in most testing machines at the peak stress. These samples, W1–5, are listed in Table 1 along with obvious optical features. Figure 2 shows the region of the complete stress–strain curve which each sample attained. All of these runs were made at pressures somewhat higher or lower than 500 bars, although at the same temperature and strain rates as our samples. Stresses are not listed in Table 1 for the W series; stress

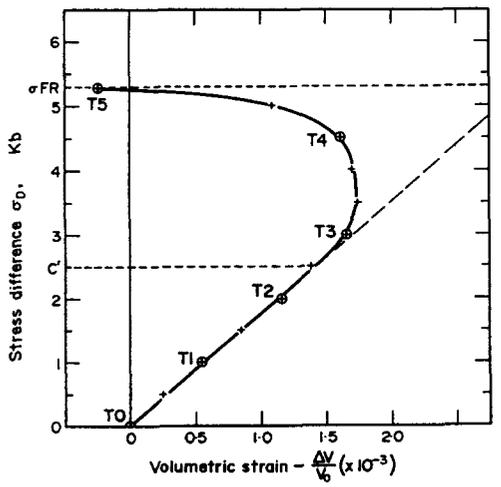


Fig. 1. Stress–volumetric strain curve for sample T5. The stresses to which the other samples were subjected are also shown from Table 1. Confining pressure was 500 bars.

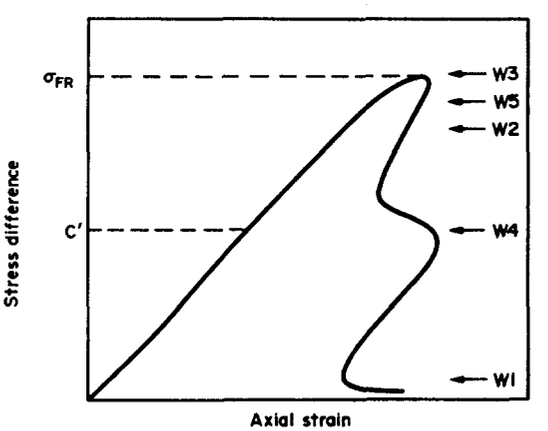


Fig. 2. Schematic stress–axial strain curve from [13] showing the stress history of the W series in Table 1.

for some of these samples could not be easily calculated as they were partially faulted.

For SEM examination a core, approx 1 cm dia, was first taken from the cylindrical triaxial sample (1·9 cm dia by 3·8 cm long) near mid-length in a direction perpendicular to the sample axis. One end of the core was ground flat and then the surface was carefully ground following the scheme in [2] to remove about 500 μm. About 20 μm more of material was then removed by ion milling and an approximately 200 Å thick layer of gold–palladium, which is the minimum thickness required for SEM operation, was applied to the surface.

Next, a photo mosaic extending over 10–30 grains was prepared with the SEM at 300–400 magnification; this is the lowest magnification at which most cracks can be detected. The mosaic served as a guide for more detailed study of individual cracks. When detailed mapping was completed the mosaic was also extremely useful in revealing spatial relationships of cracks to different minerals, as well as showing at a glance the overall orientation and distribution of stress cracks.

The absolute resolution of our SEM (JELCO Model JSM-U3 operating with an accelerating voltage of 25 kV) was around 100 Å. This was not attained owing to the metallic coating, which probably rendered features smaller than about 200 Å invisible.

OBSERVATIONS

Criteria have been suggested [2] to distinguish cracks, particularly transgranular, caused by compressive or thermal stress from crack-like, low aspect ratio cavities (LARCS) naturally present in rocks. Stress produced sharp-ended, narrow, long cracks; they lacked the round terminations and bridges often observed in natural LARCS [2, 17]. Transgranular stress cracks were particularly conspicuous [2] and, for compressive stress, were aligned nearly parallel with the maximum stress (termed here 'low angle'). These criteria were tested in the present more detailed study of stress cracking; as we will see, most were substantiated, although some new features were evident.

Stress cracks fall into two broad classes, those which are transgranular, termed type B, and those which are closely associated with pre-existing arrays of grain boundary LARCS, cleavage planes or healed cracks, called type A. For 3 samples, T0, T3 and T5, we determined crack density without regard to this grouping. We counted the number of cracks encountered in as many as 80 traverses of the sample spaced 25 μm apart, in two directions, parallel and perpendicular to stress. Results are tabulated in Table 2 and shown graphically in Fig. 3 (a and b) with arithmetic means and standard deviations.

TABLE 2. CRACK DENSITIES

Sample	Traverses					
	Direction	No.	Length mm	Total length mm	Number of crack intersections	Density mm^{-1}
T0	Radial	37	1.00	37.0	182	4.91
	Axial	41	0.90	36.9	178	4.82
T3	Radial	41	2.05	84.1	502	5.97
	Axial	83	1.00	83.0	448	5.39
T5	Radial	81	1.20	97.2	861	8.85
	Axial	51	2.05	104.5	597	5.71

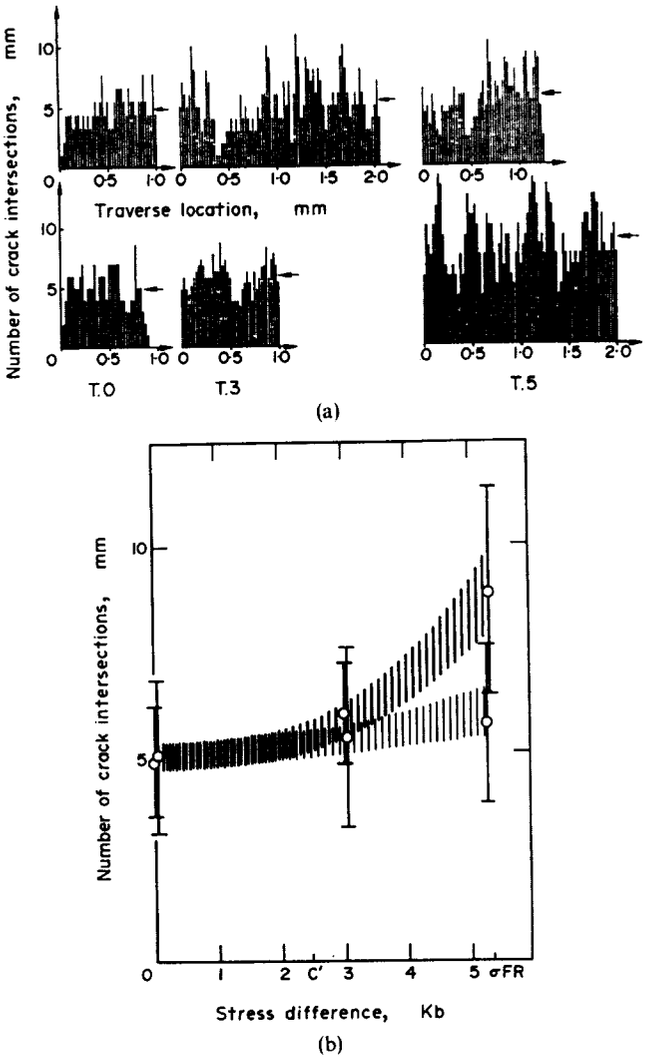


Fig. 3. Crack densities as a function of stress. (a) The lower six plots show the number of crack intersections per unit length for all traverses, for the three samples T0, T3, and T5, in axial (light) and radial (bold) traverse directions. Traverse spacing was 25 μm. Small arrows to the right of each plot show arithmetic mean density. Peaks in T5 were associated with the presence of biotite or magnetite grains. (b) Means (given in Table 2) are plotted as a function of stress with the corresponding standard deviation.

Cracks of type A

Type A cracks were in three groups, those associated with (a) arrays of natural grain boundary LARCS, (b) more or less healed transgranular stress cracks, related presumably to some early stress history, and (c) cleavage cracks.

In Westerly granite about 30% of the grain boundaries are naturally occupied by LARCS [2]. At some low stress, these LARCS begin to connect, whether by

rupture of the thin bridges which span the LARCS [1, 2, 17] or by growth of new low aspect ratio cavities. Near or beyond peak stress, in T5, W1–4, for example, most grain boundaries, whether at a high or a low angle to the stress, are almost continuously cracked. This could occur by a number of mechanisms: (a) high angle boundaries may become cracked by compressive crushing of the bridges, in the manner detailed previously [3]; (b) high angle boundaries may be pulled apart during unloading; or, (c), low angle boundaries open as the rock dilates perpendicular to the stress. Because of the complicated geometry of grain boundary cavities, a step-by-step development of this cavity enlargement is extremely difficult to trace. Thus we are not able to report with any confidence differences between T2–4. The appearance of a completely cracked boundary typical of T5, for example, is identical with that resulting from a high pressure cycle (Figs. 1 and 3) [3]. Examples from the present work are shown in Figs. 7, 9 and 11.

Healed cracks particularly associated with quartz, served here as the sites of very subtle stress effects. Three healed cracks, all at 10–30° to stress, are shown in Fig. 4. At very low stress (T0, T1) they have the appearance of long, thin, more or less completely filled openings of nearly constant cross-section; they may contain widely-spaced LARCS with rounded ends (Fig. 4a). Study of 22 of these at our highest magnification (30,000 ×) revealed no new cracks in T0 and T1. In T2 and T3 new cracks could just be detected; one is shown in Fig. 4b, which also illustrates *ressuage*, a

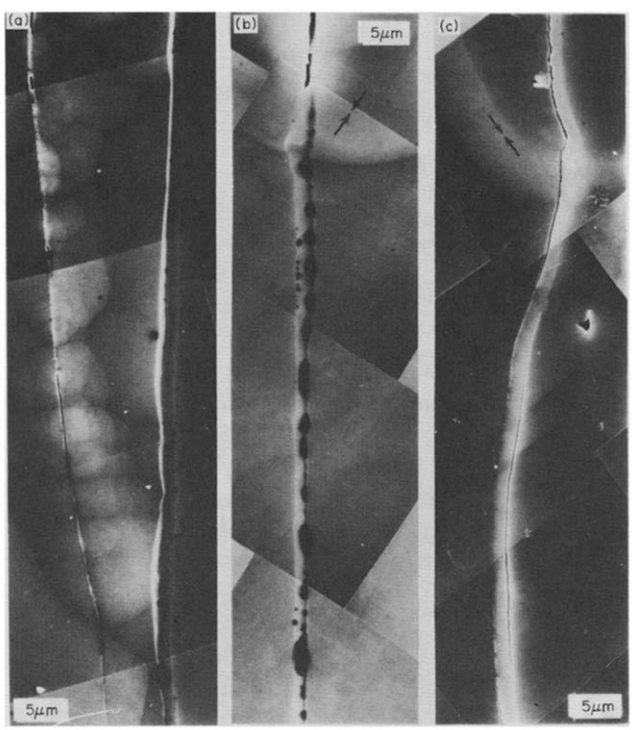


Fig. 4. Photomicrographs of healed cracks in quartz. No incipient cracking is evident in sample T0 (a) along the boundaries of quartz and healing material. A very narrow crack decorated by oil droplets was typical of T3 (b), whereas an almost continuous open crack occurs along the old crack filling in T5 (c) (right). The maximum stress direction is shown by arrows. The white bars give the scale.

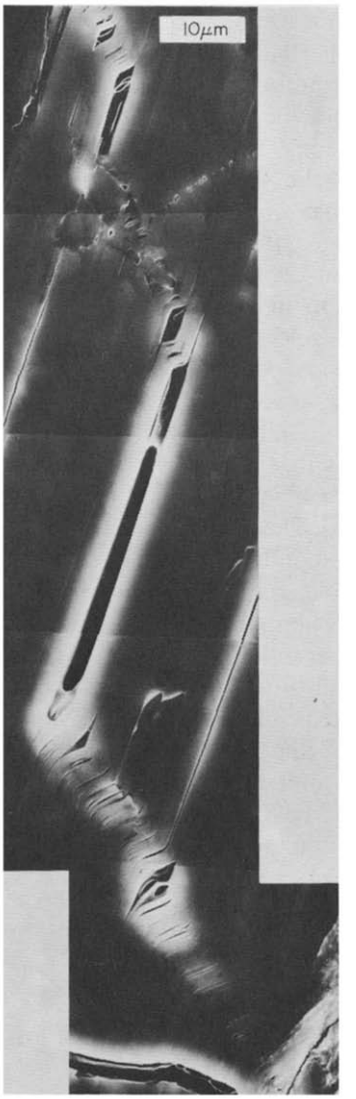


Fig. 5. Photomicrograph of a deformed biotite grain from W1 (Table 1). The long cracks are parallel with (001) and stress was vertical. At least two kink bands are evident as are the numerous holes which must have been forced open while the rock was under pressure. The viscous material which appears within several of the larger cavities is epoxy used to hold this weakened sample together prior to sectioning.

decoration technique [25]. With this technique cracks with a width below our resolution (about 0·02 μm) could be detected. In Fig. 4b a row of black globules runs along one side of an old crack filling; the filling is seen to be about 1 μm thick and of lighter color than the quartz on either side. The black blobs, also seen in Fig. 10, are drops of some viscous fluid expelled from either a closed or invisible crack which runs along the line of blobs. Their round forms, their variable size with time, their lack of a metallic coating (which makes them look black in the SEM) and the fact that they can be washed off support this contention. Because they are held by capillarity, the blobs only adhere to the finest cracks. The fluid is most likely an oil introduced during sample preparation.

The fine-scale cracking typically shown in Fig. 4b was characteristic of the two samples close to the stress C' (Fig. 1). T4 and T5 show much more extensive and

obvious cracking along old healed cracks, illustrated, for example, in Fig. 4c. Here a new crack alternates from one side of the old filling to the other; it maintains a nearly constant opening (0·1 μm) along its length. In T5 only 2 out of 17 such healed cracks which were at a low angle to stress remained intact. In all other cases, the filling was cracked completely along at least one boundary. Small shear offsets of the two sides of the filling were often evident in T5.

In Fig. 9 is an example of a high stress dilatant crack which both begins at and follows previous LARCS or healed cracks. A remarkable feature is the large constant opening of the axial in contrast to the inclined segments. We take this as evidence of sliding on the inclined segments; about 0·25 μm of shear displacement apparently occurred in this example.

Cracks parallel with cleavage were rather uncommon and limited to two minerals, biotite (at and beyond peak stress as shown in Fig. 5) and microcline. Cleavage cracks first appeared in microcline at low stress (T2); they were consistently within 10° of the stress direction.

Cracks of type B

Transgranular cracks, termed type B cracks, were of three types, those at interior pores in grains, as illustrated in Fig. 6, those starting at high angle grain boundaries of different minerals (Fig. 7), and those starting at pre-existing grain boundary LARCS which are inclined 20°–60° to compression (Fig. 8). A few have no such associations, as illustrated in Fig. 10, at least in the plane of the section.

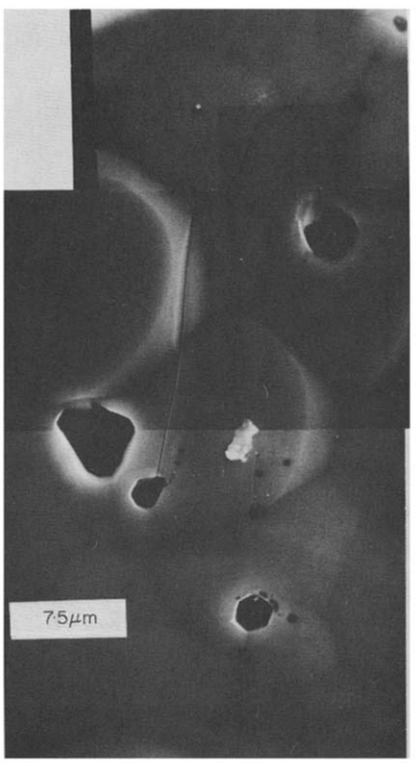


Fig. 6. A transgranular crack in magnetite in T5. The black cavities are natural pores in this mineral. Stress was vertical.

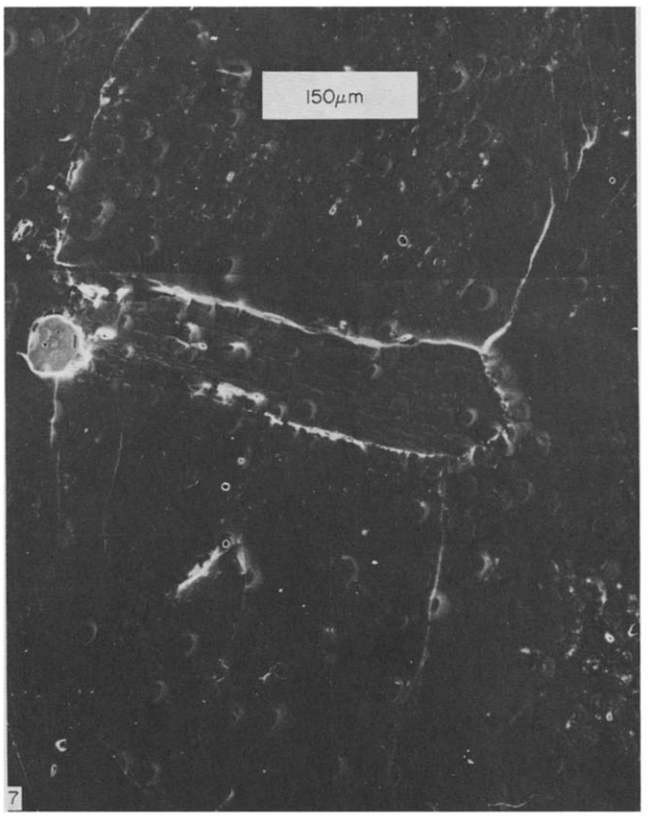


Fig. 7. Transgranular cracks at biotite grain boundaries in T5. Stress was vertical. Cleavage in biotite is parallel to long dimension of the grain. Note tendency of cracks to begin at biotite and die out in surrounding grains. Quartz is the darker phase.

Transgranular cracks were first detected in T4. Some die out with a sharp end within the grain (Fig. 6), although close to peak stress, they typically traverse the entire grain (Fig. 3) [2]. Occasionally such a crack will refract along the boundary to continue along some new low angle direction in another grain. Beyond peak stress, cracks may extend across several grains, as noted optically [13]; this produced cracks up to 500 μm long in T5 at peak stress. With the typical crack width of 0·1 μm, aspect ratio would be about 10^{-4} in this sample while it is being viewed at atmospheric pressure.

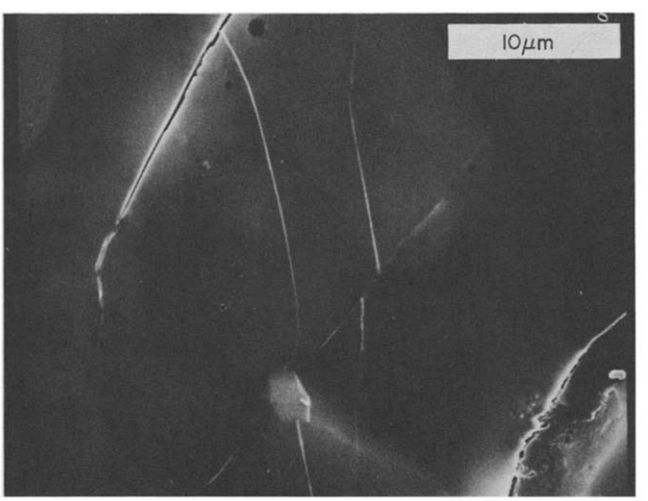


Fig. 8. Association of new stress cracks with natural LARCS which were inclined to the stress (vertical). Sample T5.

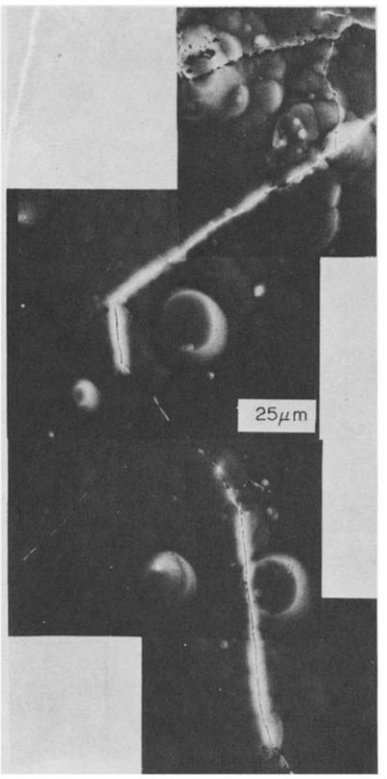


Fig. 9. Dilatant cracks associated with inclined pre-existing LARCS in T4. Note that the opening of the two lower vertical cracks appears to remain constant from one shear crack to another. Stress was vertical.

Cracks emanating from a high angle interface of two different minerals were surprisingly common in T4, T5 and all the W series. Biotite was often involved, as shown in Fig. 7; here, cracks spread outward into surrounding quartz and feldspar.

The final intragranular crack association was rather uncommon, contrary to expectation, being limited to rare situations at inclined LARCS (Fig. 8), or biotite grains where slip occurred on (001) planes (Fig. 11). For example, only one clear example of grain boundary sliding was found in the T series (Fig. 9).

At, and beyond, peak stress, biotite deformation and the cracking in nearby grains related to it was striking (Fig. 11). For example, in T5, 20 out of 28 grains of biotite appeared to be the source of transgranular cracks in nearby quartz and feldspar. These cracks had aspect ratios of 10^{-3}–10^{-4}. Some appeared to follow healed fractures.

Kinking in suitably oriented biotite grains (Figs. 5 and 11) was evident in all of the post-peak specimens up to a pressure of 1·5 kbar. It produced cavities in the biotite resembling those shown in Fig. 5. The abundant cracks just noted above, all nearly parallel with compression, are most conspicuous adjacent to grains of biotite which are most strongly kinked (Fig. 11).

Magnetite, another constituent in Westerly granite, was conspicuously cracked (Figs. 6, 10 and 11), starting fairly early in the stress history. At peak stress nearly all grains contain new transgranular cracks. Among the major phases, quartz and microcline contain more cracks than plagioclase, up to peak stress.

Summary of crack development prior to faulting

Our SEM studies suggest several distinct stages in crack development leading up to the beginning of faulting in Westerly granite at 500 bars confining pressure. These are shown diagrammatically in Fig. 12 compared with a complete stress–strain curve typical of granite at this pressure [13].

(a) Up to 1–2 kbars stress, no stress cracks are formed.

(b) Between 2 and 3 kbars, near the stress C', fine cracks of type A open along grain boundaries or healed cracks oriented up to 30° to compression.

(c) At around 4 kbars, type B cracks form, starting at high angle grain interfaces, or rarely, at inclined LARCS.

(d) Growth of both A and B types accelerates at and above 5 kbars, probably accompanied by gliding flow in suitably oriented biotite.

(e) At, and beyond, peak stress, kinking occurs in all suitably oriented biotite, producing conspicuous open cavities. Long cracks extend from the biotite into adjoining quartz and feldspar. Often the longest and most conspicuous cracks just prior to faulting are those associated with biotite.

DISCUSSION

Recognition of new stress cracks

The criteria suggested previously [2] for the distinction between pre-existing and new crack-like cavities seem in general verified by our present study. In addition, we noted that stress cracks have sharp rims regardless of the phase involved and are of nearly constant width for most of their length. One difficulty with both pre-existing and new features is the limited resolution of the SEM. Some new, sharp-ended cracks may be present in our low-stress samples but not visible because they are below our resolution. Even unstressed

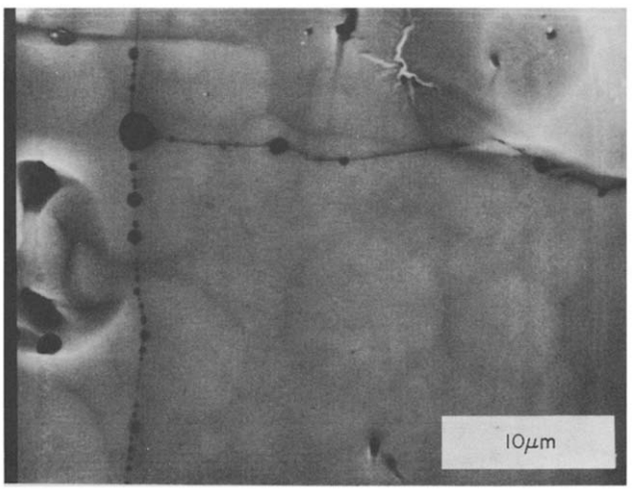


Fig. 10. Example of a very narrow stress crack (vertical, marked by oil droplets) unrelated to pre-existing cavities in T5. It intersects a stress crack in an abnormal orientation (horizontal, also marked by oil droplets). The latter was formed later during unloading. Stress was vertical.

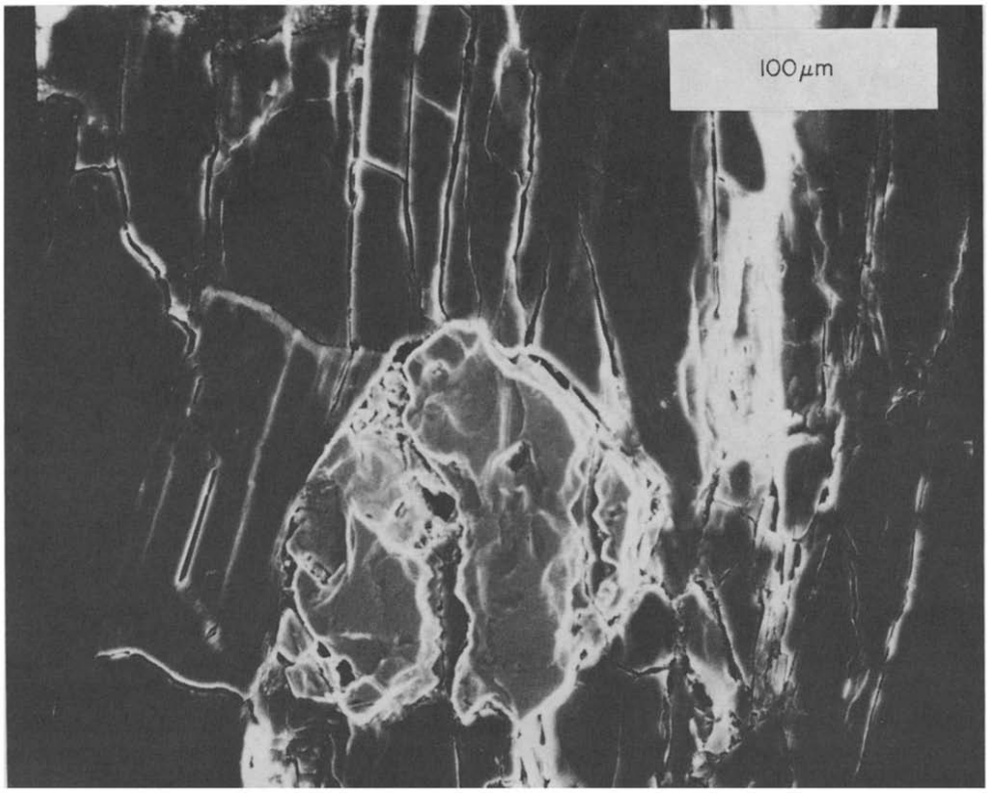


Fig. 11. Heavily cracked magnetite at low magnification in W1. To the left is same biotite grain as Fig. 5. Long cracks extend outward from both magnetite and ends of kink bands in biotite. All grain boundaries in picture are cracked. Note (sub-horizontal) unloading cracks in upper field. Stress was vertical.

natural material may have a somewhat different aspect at a still higher magnification, although the widespread healing of old cracks evident in many granites [2, 17] ought to have eliminated most very narrow cavities. In any case, this is clearly a direction in which further work is required. Additional magnification, without the added disadvantage of a metallic coating, could be gained by transmission microscopy either of foils, or replicas made from ion-thinned surfaces.

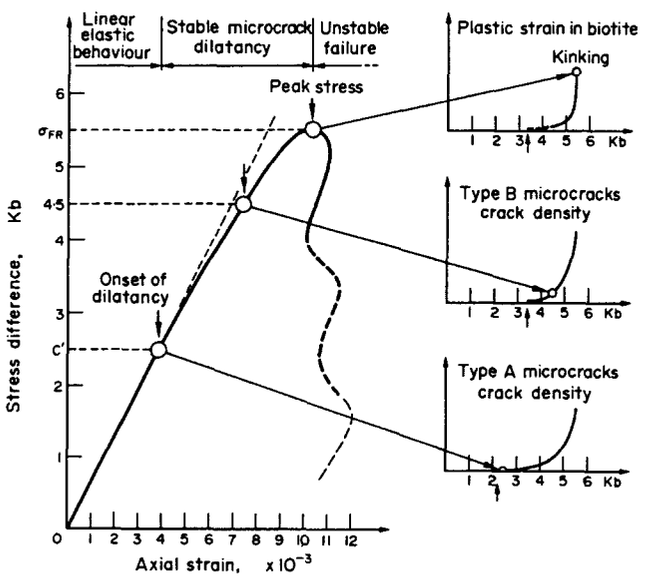


Fig. 12. Idealized stages in crack development, shown relative to a schematic stress–strain curve. Small graphs on right show qualitatively how density of cracks and plastic strain develops with stress.

The onset of dilatancy

The first appearance of new, stress-related, cracks in T2 seems to agree fairly well with the onset of dilatancy, C', determined by volumetric strain measurements (Fig. 1). However, because of our rather wide spacing in stress, and difficulties inherent in picking a gradual break in slope [4], uncertainty in both methods is at least 500 bars, or about 10% of the peak stress. As new cracks are very small, as seen in T2 for example, it may be difficult to reduce this uncertainty with the SEM.

Crack density and stress

We can directly compare our crack densities with those determined optically for Westerly granite [13]. Thus, near peak stress crack density (last column in Table 2) in the radial direction nearly doubled; the optical measurement gave nearly a three-fold increase (Fig. 6 of [13]). The most likely source of this discrepancy was the inability to observe original cracks optically. Our density increase is very close to that reported both for uniaxial and triaxial samples of Chelmsford granite [14], a coarser-grained rock. A much greater increase in crack density, nearly a factor of 10, was reported for a micaceous quartzite [15]. This could be typical of micaceous rocks, or be due to the particular way in which confinement was allowed to vary in these triaxial tests.

Our value of crack density is roughly in accord with lateral strain. For a density of 10/mm and a typical

opening of 0·1 μm, the nonelastic radial strain [4], ϵ'_r would be 1×10^{-3}. We observed 2×10^{-3}.

Relation of dilatant cracks to shear

Dilatant cracks were thought on the basis of strain measurements [4] to have one of two forms, shown in Fig. 13. One was the isolated crack (a) and the other (b) was supposed to form at the ends of one or two *shear cracks*, that is, pre-existing LARCS oriented so that shear could occur. All new cracks were supposed to be approximately parallel with the maximum compression. The form shown in (b) would account for the hysteresis observed [11, 19–21] when a rock is cycled in the dilatant region.

Although consistent with strain measurements, not all these suggestions agree with our present observations. The most common forms observed in the SEM were (a) cracks which connected rows of low angle grain boundary LARCS (Fig. 4) and (b) transgranular cracks which intersected high angle interfaces of dissimilar minerals (Fig. 7). Only a small proportion (about 5%) terminated in even a single shear crack. Perhaps (a) is the isolated crack of Fig. 13a, but shear-related features in the form anticipated (Fig. 13b) were insignificant. This raises doubts about the current explanations for hysteresis. We return to this question below.

Others have also had difficulty finding shear-related cracks. In several optical studies of rocks stressed to near peak stress none were reported [11, 12, 14, 15], although Friedman *et al.* [12] showed an array of shear cracks which apparently developed into a fault. For the post-peak region, Wawersik & Brace [13] reported that shear cracks themselves became the dominant mode and that they extended in their own direction. Our re-examination of the W series with the SEM showed that these so-called 'shear cracks' may actually consist of axial arrays. All others have noted widespread transgranular cracking in both pre- and post-peak regions, and Hallbauer *et al.* [15] observed interface-related cracks in a quartzite. Of course, the apparent absence of shear-related cracks as seen optically could be explained by their small size. As noted above, the shear cracks are especially narrow and usually occur at a grain interface; this could render them nearly invisible optically.

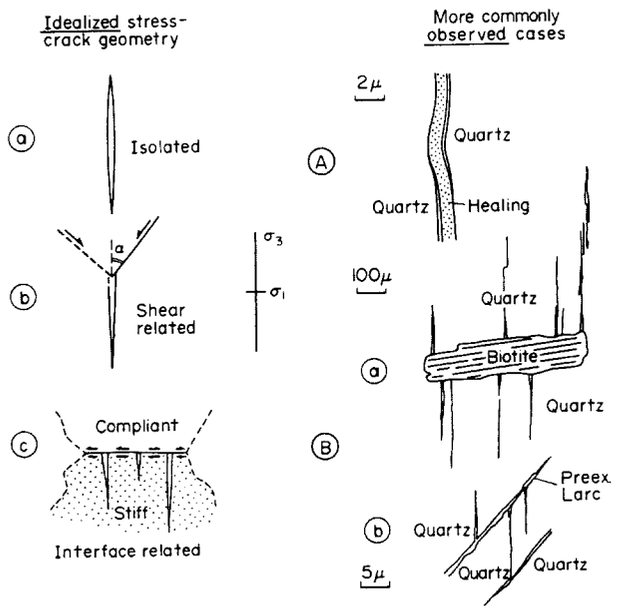


Fig. 13. Idealized and observed stress crack association. Sketches on left show the prime modes of cracking. Sketches on right are idealized from Figs. 4, 7 and 8 respectively. 2a, on right, was the most commonly observed. Cross, center, gives stress directions.

An independent set of observations seems consistent with the scarcity of shear-related cracks. Consider the linear strains, ϵ_z and ϵ_r, measured during a stress cycle. Comparison of the non-elastic parts of the axial and radial strains, ϵ'_z and ϵ'_r, [after 4] shows that $\epsilon'_r/\epsilon'_z \approx \tan \alpha$, where α is the inclination of the shear feature to the direction of maximum compression. For cracks so oriented that shear is likely, $\alpha = 30$–$40°$ so that ϵ'_r/ϵ'_z should be 2 or less. When strains are observed at pressures of several kbars, the ratio does fall approximately in this range [4, 20], but at lower pressures, where the observed ratio is much higher, shear-related dilatant cracks, just on this basis alone seems unlikely.

Either isolated or interface-related cracks (Fig. 13a or c) clearly might account for both observed dilatancy and high ratios of ϵ'_r/ϵ'_z, although as we suggest below, only the latter could be responsible for hysteresis. To what are interface-related cracks due? This question has been widely addressed, both from the point of view of understanding thermal cracking [23] and in studies of the rock–platen interface in rock compression testing [most recently 14, 24]. The cause of the cracks sketched in Fig. 13c could be local tensile stress [22] at the interface of two solids of different elastic or plastic properties. In the elastic case, tension is approximately proportional to the difference in ν/E, where ν is Poisson's ratio and E is Young's modulus; tension and, therefore, cracks occur on the stiff side where ν/E is lower. According to one analysis, radial tension may reach 20% of the applied compressive stress, for a difference of ν/E of only a factor of 10 [24].

Hysteresis in dilatancy

Usually, hysteresis in stress–strain curves is attributed to sliding against friction on suitably oriented surfaces. Absence of the shear-related cracks, both in our SEM and in previous optical studies, therefore seems surprising in the light of the hysteresis so widely observed [4, 11, 19–21] when rock is cycled in the dilatant region. One possibility is that shear cracks are actually abundant, but are so narrow they fall below the resolution even of the SEM. For example, the shear cracks shown in Figs. 8 and 9 could have easily been overlooked. This point, too, might be resolved by microscopy (replica or transmission) at still higher magnification.

One source of hysteresis could be the interface-related cracks. Again, according to mathematical analysis of a rock–platen interface [14, 24], shear stress is developed at the interface as suggested in Fig. 13c, for the

particular arrangement of stiff and compliant members shown. From one analysis [24] this shear stress may reach 30% of the applied normal stress. This would probably not be sufficient to cause frictional sliding of a completely sealed grain interface but might be adequate when the interface is the site of pre-existing LARCS as in Fig. 7.

Another source of hysteresis may be the biotite. As noted above, slip on (001) becomes pronounced near peak stress for favorably oriented grains; perhaps it begins at much lower stress. In any event, as this mineral is a minor constituent, it probably plays a contributory role at best.

Still another source of hysteresis may be the axial cracks themselves. In the first place, crack growth in itself is an energy-dissipating process. Nevertheless an open axial crack of elliptical form, oriented exactly parallel with stress, ought to contribute little to the energy lost due to friction. However, this idealization may rarely hold. Hysteresis may arise from several situations in which shear stress is transmitted across axial cracks, if, for example (a) the crack is inclined to the stress and, therefore, not truly 'axial' (Fig. 4), or (b) the crack edges are rough and somewhat interpenetrating, or (c) the cracks are closed and situated at interfaces between different minerals. The common situation of unloading cracks, between two axial cracks (Fig. 10), is most easily explained by the existence of locked-in shear stress on axial cracks.

Thus, a number of mechanisms are available which might explain in a qualitative way the observed hysteresis in dilatancy, in spite of the apparent scarcity of shear cracks. Just which of the above mechanisms dominates may become clear as regions of the stress–strain curve are studied with respect to crack development in even greater detail. Perhaps analysis of the shapes of the stress–strain loops themselves may eliminate several of the above mechanisms.

The brittle strength of Westerly granite

Plastic flow of biotite (Fig. 5) may to some extent limit the strength of Westerly granite. The close association of biotite grains with zones of intense cracking and even with faults themselves has been noted elsewhere [11, 12]; kinking has also been described elsewhere from post-peak uniaxial experiments [12]. It is clear in our samples that the plastic slip in biotite caused the cracking in nearby grains. Hence, perhaps it is the kinking, itself an unstable process, which determines the instability of fracture, rather than, say, the stability of the tiny columns of intact rock between dilatant microcracks [14]. If true, this could be an important clue towards understanding what determines brittle strength, not only in biotite granite but in a wide variety of rocks. Minor constituents, either capable of flowing plastically at modest temperatures, or which have abnormally high (like magnetite) or low (like biotite along [001]) elastic moduli may play as important a role as the more obvious parameters such as major mineralogy, grain size and porosity. Clearly a series of experiments to explore this potential role of minor constituents would be of great fundamental interest.

CONCLUSIONS

(1) Most stress cracks stand out clearly in contrast to naturally existing crack-like cavities (LARCS) in Westerly granite. Particularly diagnostic are shape, as well as orientation of new cracks relative to stress.

(2) New cracks first begin to appear within about 500 bars of the stress C' determined by strain measurements; they appear at pre-existing intergranular and healed transgranular LARCS. At about 75% of the peak stress, new transgranular cracks appear, starting particularly at high angle interfaces of dissimilar minerals or, rarely, at inclined pre-existing LARCS on which shear has taken place.

(3) Crack density, the number of cracks per unit length in a direction normal to stress, approximately doubles up to peak stress.

(4) Peak stress coincides approximately with the onset of kinking in biotite and may be partly determined by the unstable behavior of this minor component.

(5) Dilatant cracks associated with shear features are rarely seen. Far more common is the association with grain interfaces at a high angle to the stress. Mathematical analysis of interfaces across which elastic properties change suggests that in our case the interfaces could have caused both the dilatant cracks and hysteresis.

(6) Interface-related cracks are independently suggested by the ratio of non-elastic axial and radial strains during dilatancy, particularly at low pressure.

(7) Selected microscopy at higher resolution may be worthwhile, particularly to detect the beginning of slip in biotite grains, as well as to discover any shear-related cracks too narrow to be visible with the SEM.

(8) Minerals which amount to only 5–10% by volume may limit strength in a brittle rock, if they (a) are capable of plastic slip or (b) have unusually high or unusually low elastic moduli relative to other minerals in the rock.

Acknowledgements—This research was supported by the Army Research Office, Durham, under Contract No. DAHC04-73-C-0017, and by the U.S. Geological Survey under Contract No. 14-08-0001-14551. Paul Tapponnier was supported at MIT on a grant from the French Ministry of Foreign Affairs.

Received 22 *July* 1975.

REFERENCES

1. Brace W. F., Silver E., Hadley K. & Goetze C. A closer look at cracks and pores. *Science* **178**, 162–163 (1972).
2. Sprunt E. & Brace W. F. Direct observation of microcavities in crystalline rocks. *Int. J. Rock Mech. Min. Sci. & Geomech. Abstr.* **11**, 139–150 (1974).
3. Sprunt E. & Brace W. F. Some permanent structural changes in rocks due to pressure and temperature. *Proc., 3rd Int. Congr. Rock Mech., Denver*, Vol. 2, Part A, pp. 524–529 (1974).

4. Brace W. F., Paulding B. & Scholz C. H. Dilatancy in the fracture of crystalline rocks. *J. Geophys. Res.* **71**, 3939–3954 (1966).
5. Scholz C. H., Sykes L. R. & Aggarwal Y. P. Earthquake prediction, a physical basis. *Science* **181**, 803–810 (1973).
6. Mjachkin V. I., Brace W. F., Sobolev G. A. & Dieterich J. H. Two models for earthquake forerunners. *PAGEOPH*, **113**, 169–181 (1975).
7. Ide J. M. The velocity of sound in rocks and glasses as a function of temperature. *J. Geol.* **45**, 689–699 (1937).
8. Matsushima S. Variation of the elastic wave velocity of rocks in the process of deformation and fracture under high pressure, *Disaster Prevention Res. Inst., Kyoto Univ., Bull.* **32**, 1–8 (1960).
9. Nur A. & Simmons G. Stress-induced velocity anisotropy in rock: an experimental study. *J. Geophys. Res.* **74**, p. 6667 (1969).
10. Brace W. F. & Orange A. S. Electrical resistivity changes in saturated rocks during fracture and frictional sliding. *J. Geophys. Res.* **73**, 1433–1445 (1968).
11. Paulding B. W., Jr. Crack growth during brittle fracture in compression, MIT PhD Thesis, 184 pp. June (1965).
12. Friedman M., Perkins R. D. & Green S. J. Observation of brittle deformation features at the maximum stress of Westerly granite and Solenhofen limestone, *Int. J. Rock Mech. Min. Sci.* **7**, 297–306 (1970).
13. Wawersik W. R. & Brace W. F. Post-failure behavior of granite and diabase. *Rock Mech.* **3**, 61–85 (1971).
14. Peng S. & Johnson A. M. Crack growth and faulting in cylindrical specimens of Chelmsford granite. *Int. J. Rock Mech. Min. Sci.* **9**, 37–86 (1972).
15. Hallbauer D. K., Wagner H. & Cook N. G. W. Some observations concerning the microscopic and mechanical behavior of quartzite specimens in stiff triaxial compression tests. *Int. J. Rock Mech. Min. Sci. & Geomech. Abstr.* **10**, 713–726 (1973).
16. Mogi K. Some precise measurements of fracture strength of rocks under uniform compressive stress. *Rock Mech. Eng. Geol.* **4**, 41–55 (1966).
17. Montgomery C. W. & Brace W. F. Micropores in plagioclase. *Contrib. Mineral. Petrol.* **52**, 17–28 (1975).
18. Brace W. F. & Bombolakis E. G. A note on brittle crack growth in compression. *J. Geophys. Res.* **68**, 3709–3713 (1963).
19. Scholz C. H. & Kranz R. Notes on dilatancy recovery. *J. Geophys. Res.* **79**, 2132–2135 (1974).
20. Zoback M. D. & Byerlee J. D. The effect of cyclic differential stress on dilatancy in Westerly granite under uniaxial and triaxial conditions. *J. Geophys. Res.* **80**, 1526–1530 (1975).
21. Hadley K. V_p/V_s variations in dilatant rocks. *PAGEOPH*, **113**, 1–23 (1975).
22. Lisowski A. Failure types of rock cubic specimens in the light of the theory of elasticity. *Bull. de l'Academie Polonaise des Sciences*, **7**, 341–351 (1959).
23. Richter D. & Simmons G. Thermal expansion behavior of igneous rocks. *Trans. Am. Geophys. Union*, **55**, p. 423 (1974).
24. Brady B. T. Effects of inserts on the elastic behavior of cylindrical materials loaded between rough end-plates. *Int. J. Rock Mech. Min. Sci.* **8**, 357–369 (1971).
25. Leiris H. de Les formes modernes de l'essai de ressuage. *Bull. Assoc. Techn. Marit. Aeronaut.* **50**, 393–403 (1951).
26. Alburger J. R. Theory and applications of liquid tracers *Non-Destructive Testing*, **20**, pp. 91–97 (1962).